



NetSys Mapper

Intro to Cyber + Linux Fundamentals Project

NetSys Mapper is a three-phased reconnaissance mission.

Phase 1, "Network Mapping," commands a detailed survey of the internal network terrain, identifying all devices, their communication protocols, and strategic infrastructure points.

Phase 2, "External Intel Gathering," deploys digital surveillance via Shodan and WHOIS, examining the network's public presence and analysing traffic for operational security. Execute with precision to secure a comprehensive battlefield overview.

Phase 3, "Gather System Info." This task involves creating a smart script to dig into the system and pull-out key details. You'll need to find out network info like IP and MAC addresses, check how the system is running by looking at CPU and memory use, and figure out which files and folders are the biggest. Your mission is to do this neatly and efficiently, giving us a clear picture of what's going on inside the system.

Project Structure:

Phase 1

1. Map the Network - Present a network map showing ALL the devices in your home network.

1.1. Display Devices IP Address.

1.2. Display Devices MAC Address and Vendor (First 3 blocks, e.g. 12:4F:C2:XX:XX:XX).

1.3. Display the Router's Internal and External IP Addresses (Last 1 block, e.g. XXX.XXX.XXX.123).

1.3.1. Use [] to wrap around each . (e.g. 10[.]123[.]123[.]123)

1.4. Display Device Names.

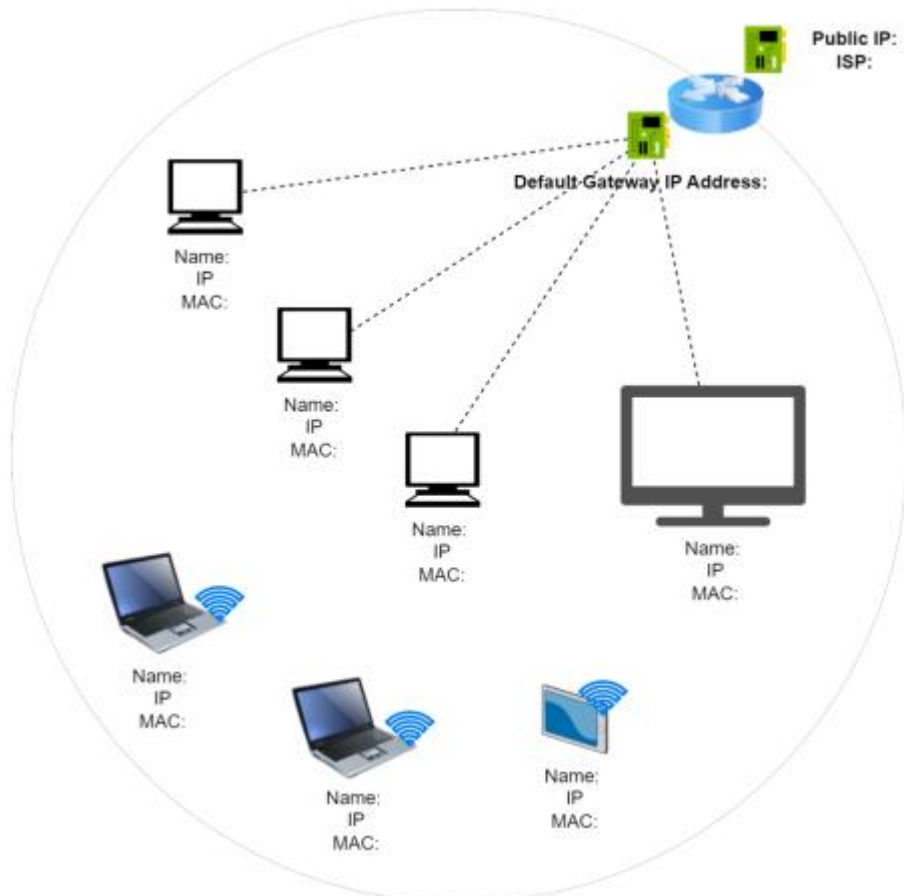
1.4. Display the DNS and DHCP IP Addresses in your Network.

1.5. Display your Internet Service Provider (ISP).

1.6. Display if the Device is Connected via Ethernet or Wireless.

1.7 Display the Operating system (OS) and version of your devices.

Sample Network Map



Phase 2: Collecting Information

- 2.1. Use Shodan to Check Your Public IP Address.
- 2.2. Use WHOIS to Check Who is Registered on Your Public IP Address.
- 2.3. Sniff Your Network and Identify Three (3) Used Protocols.
 - 2.3.1. For Each Protocol Explain Its Usage.
 - 2.3.2. For Each Protocol, Find the Used Port Number.

General Websites available: shodan.io, viewdns.info, draw.io

Phase 3: Automated System Info Extractor

Create a bash script that does the following:

- 3.1. Identify the system's public IP.
- 3.2. Identify the private IP address assigned to the system's network interface.
- 3.3. Display the MAC address (masking sensitive portions for security).
- 3.4. Display the percentage of CPU usage for the top 5 processes.
- 3.5. Display memory usage statistics: total and available memory.
- 3.6. List active system services with their status.
- 3.7. Locate the Top 10 Largest Files in /home.

General Tools and Commands: curl, ifconfig/ip addr, top, ps, du, find, standard Bash scripting utilities.